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MILITARY SPECIFICATION

AMPLIFIER-POWER SUPPLY GROUP OA-3633( )/GRC

1. SCOPE

1.1 This specification covers one type of vehicular amplifier-power supply designated as Amplifier-Power Supply Group OA-3633( )/GRC which is used for vehicular operation of Receiver-Transmitter, Radio RT-505( )/PRC-25. The group consists of Amplifier-Power Supply AM-2060( )/GRC and Cable Assembly, Special Purpose, Electrical CX-4655( )/GRC. (See 6.1 and 6.3).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids (or request for proposal), form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-636	Box, Fiberboard.
PPP-B-640	Boxes, Fiberboard, Corrugated, Triple-Wall
PPP-F-320	Fiberboard; Sheet, Stock and Cut Shapes
PPP-T-76	Tape; Pressure-Sensitive, Adhesive, Paper Water Resistant
PPP-T-97	Tape; Pressure-Sensitive, Filament Reinforced

MILITARY

MIL-P-116	Preservation, Methods of
MIL-B-117	Bags, Interior Packaging
MIL-B-121	Barrier Material, Greaseproofed Flexible (Waterproofed)
MIL-S-901	Shockproof Equipment, Class HI (High-Impact), Shipboard

FSC 5820

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SPECIFICATIONS

MILITARY (Cont'd)

MIL-P-11268	Parts, Materials, and Processes Used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-B-43291	Box, Fiberboard, Corrugated, Double-Wall, Weather Resistant
MIL-P-55110	Printed Wiring Boards

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection By Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized Unit Loads, 40 inches X 48 inches; 4-way Partial and 4-Way Pallets
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-810	Environmental Test Methods.

DRAWINGS

ELECTRONICS COMMAND

SC-A-46439	List of Accessories for Package Tester.
SC-GL-57992	Drawing and Gage List for Amplifier-Power Supply Group OA-3633( )/GRC.
SC-DL-454800	Drawing and Data List for Amplifier-Power Supply AM-2060( )/GRC
SM-D-454880	Cable Assembly, Special Purpose, Electrical CX-4655( )/GRC.

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(Copies of specification, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Description.- Amplifier-Power Supply Group OA-3633( )/GRC furnishes the regulated direct-current (dc) power required for vehicular operation of Radio Sets AN/VRC-53( ), AN/VRC-64( ), AN/GRC-125( ) and AN/GRC-160( ). The amplifier provides the amplification necessary to drive a self-contained panel-mounted loudspeaker. In addition, provisions for mounting of Receiver-Transmitter Radio RT-505( )/PRC-25 and RT-841( )/PRC-77 are included. Cable Assembly, Special Purpose, Electrical CX-4655( )/GRC provides for the interconnection of the receiver-transmitter to Amplifier-Power Supply AM-2060( )/GRC.

3.2 First Article Samples.- The contractor shall furnish four (4) first article samples for approval, if required by the invitation for bid and contract. (See 4.3).

3.3 Construction.- Amplifier-Power Supply Group OA-3633( )/GRC shall be constructed in accordance with the requirements of this specification and of the drawings listed below (including the requirements for parts, materials, and processes thereon).

Unit	Drawing
AM-2060( )/GRC	SC-DL-454800
CX-4655( )/GRC	SM-D-454880

3.4 Printed Wiring Assemblies.- Materials and processes for printed wiring assemblies shall conform to Specification MIL-P-55110 unless otherwise specified in the equipment drawings (See 4.4).

3.5 Parts, Materials and Process; General.- In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein, shall apply.

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3.6 Finish.- The equipment shall be finished in accordance with Specification MIL-F-14072 and the equipment drawings. (See 4.4) The final paint film on Type I surfaces shall be green color (olive-drab) enamel matching a color chip provided by the procuring agency. (See 6.6)

3.7 Marking.-

3.7.1 General.- Marking shall conform to Specification MIL-M-13231. (See 4.4) Panel markings shall be Group I as described in that specification and the equipment drawings.

3.7.2 Visibility.- Wherever p.acticable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.7.3 Serial numbers.- Amplifier-Power Supply AM-2060( )/GRC shall have serial numbers.

3.8 Service conditions.- The equipment shall meet the following service conditions:

3.8.1 Operation.- Continuous use for a period of one year, under the environmental requirements specified in 3.8.2 through 3.8.8.2, with a duty cycle of 23 hours on and 1 hour off with no more than normal maintenance and replacement of parts. Where the contractor is required to make a selection of parts, materials, processes, construction methods, etc., he shall be guided by this requirement. Approval of the first article sample (see 3.2) shall be considered as compliance with this requirement.

3.8.2 Temperature.-

3.8.2.1 Low.- During testing as specified in 4.7.1.1, the equipment shall meet the following requirements:

Paragraph	Measurement
3.9.1	Output Voltage
3.9.2	Input Current
3.9.3	Audio Output
3.9.4	Audio Distortion
3.9.5	Antenna Frequency Selector Control
3.9.7	Audio Response
3.9.8	Input Voltage Range

After testing, the equipment shall meet full specification performance for these measurements.

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3.8.2.2 High.- During testing as specified in 4.7.1.2, the equipment shall meet the following requirements:

Paragraph	Measurement
3.9.1	Output Voltage
3.9.2	Input Current
3.9.3	Audio Output
3.9.4	Audio Distortion
3.9.5	Antenna Frequency Selector Control
3.9.7	Audio Response
3.9.8	Input Voltage Range

After testing, the equipment shall meet full specification performance for these measurements.

3.8.3 Humidity (moisture-resistance).- During testing as specified in 4.7.2, the equipment shall meet the following requirements:

Paragraph	Measurement
3.9.1	Output Voltage
3.9.2	Input Current
3.9.3	Audio Output
3.9.4	Audio Distortion
3.9.5	Antenna Frequency Selector Control
3.9.7	Audio Response
3.9.8	Input Voltage Range

After testing, the equipment shall meet full specification performance for these measurements.

3.8.4 Altitude (elevation).- The equipment shall meet full specification performance for the following measurements both during and after testing as specified in 4.7.3.

Paragraph	Measurement
3.12	Operational Inspection

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3.8.5 Leakage (immersion).- The equipment shall show no evidence of leakage after it is tested as specified in 4.7.4.

3.8.6 Dust.- After testing as specified in 4.7.5, the equipment shall meet full specification performance for the following measurements:

Paragraph	Measurement
3.12	Operational Inspection

3.8.7 Salt fog.- After testing as specified in 4.16, the equipment shall show no evidence of corrosion as evidenced by any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface.

3.8.8 Vibration, bounce, and shock.-

3.8.8.1 Vibration, internal.- The amplitude of vibration of any part, subassembly or structural member of the equipment shall not exceed twice the amplitude of the vibration applied to the equipment at any frequency between 10 and 55 Hertz (H_z). (See 4.7.6)

3.8.8.2 Bounce and shock.- The equipment shall be capable of meeting the requirements of Table I.

Table I.- Bounce and Shock

Test	Para No.	Performance after test
Bounce	4.7.7	Specified performance. (Note A) No physical damage.
Shock, ballistic	4.7.8	Operational. (Note A) Any physical damage shall be minor only.

NOTE A: The amplifier-power supply shall meet specified performance for the following requirements: 3.9.1, 3.9.3, 3.9.4, 3.9.5 and 3.9.7.

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3.9 Electrical Requirements, Amplifier-Power Supply AM-2060( )/GRC.-

3.9.1 Output voltage (See 4.8.2).- The output voltage shall be within the limits specified below when the amplifier-power supply is loaded with Receiver-Transmitter, Radio RT-505( )/PRC-25, RT-841( )/PRC-77 or equivalent.

Voltage	Voltage with 22.0 vdc input	Voltage with 26.0 vdc input	Voltage with 30.0 vdc input
B1	13 ±0.5	13 ±0.3	13 ±0.5
A1	2.00 min.	2.78 ±0.12	3.5 max.

The B1 output voltage shall not vary more than 0.5 volt between receive and transmit loads.

3.9.2 Input current (See 4.8.3).- The input current, under the load conditions of 4.8.1 shall not exceed the following:

RT-505( )/PRC-25 Load Condition	Maximum Current (Amperes)
Receive	0.5
Transmit	1.0

3.9.3 Audio output (See 4.8.4).- The audio output shall not be less than 1.0 watt with 3.0 milliwatts input at 1000 H_z.

3.9.4 Audio distortion (See 4.8.5).- The harmonic distortion shall not be greater than 7 percent.

3.9.5 Antenna frequency selector control (See 4.8.6).- The antenna frequency selector control provides the switching information to the vehicular antenna matching unit network. A ten position control is provided with the lowest frequency position at the extreme counter-clockwise setting. When measured in accordance with 4.8.6, 26 ±1 vdc shall be provided across the applicable pins of the antenna control receptacle connector located on Amplifier-Power Supply AM-2060( )/GRC.

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3.9.6 No load voltage (See 4.8.7).- The no load voltage shall not exceed the following with 30.0 volts dc input and all loads simultaneously removed.

A1	Input voltage ±0.5 vdc
B1	15.0 vdc

3.9.7 Audio response (4.8.8).- The audio response of the amplifier shall be within the following limits:

Audio Frequency (H_z)	Audio response
1000	0 db = 1.0 watt (reference)
150	+1 db, -3 db
300	±1.0 db
2500	±1.0 db
3000	±3.0 db
5000	+1 db, -3 db

3.9.8 Input voltage range (See 4.8.9).- The amplifier-power supply shall meet full specification requirements when operating from a 22.0 to 30.0 vdc vehicular battery source. Nominal input voltage is 26.0 vdc.

3.10 Preconditioning (See 4.6).-

3.10.1 Bounce preconditioning.- The equipment shall be capable of meeting the requirements herein, without subsequent processing, after subjection to the bounce preconditioning of 4.6.1. (See also 4.5).

3.10.2 Electrical Preconditioning.- The equipment shall be capable of meeting the requirements herein, without subsequent processing after subjection to the electrical preconditioning of 4.6.2. (See also 4.5)

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3.11 Air Seal Test, Provision for (See 4.11).- Amplifier Power Supply enclosure shall provide means for enabling performance of the air seal test.

3.12 Operational inspection.- The operational test specified in 4.12 shall be performed to determine that the equipment is operable prior to packaging and at such other times as required.

3.13 Interchangeability.- Like units, assemblies, subassemblies and replaceable parts shall conform to requirement 7 of MIL-STD-454 and shall be physically and functionally interchangeable, without modification of such items or of the equipment. (See 4.9)

3.14 Electromagnetic interference characteristics.- The equipment shall meet the electromagnetic interference requirements specified in MIL-STD-461 for Class I. Tests and test methods shall be as specified in MIL-STD-462. If the contractor considers it desirable to modify the equipment proper in order to facilitate conformance to the specified test limits, he shall obtain approval from the contracting officer before proceeding with the modification.

3.15 Technical literature, tools, and running spare parts.- Technical literature, tools and running spare parts shall be furnished if specified in the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order. (See 6.2)

3.16 Fungus.- The equipment (including accessories) shall show no evidence of viable fungus or corrosion when subjected to the test specified in 4.15. Corrosion is any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface.

3.17 Workmanship.- The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs in Specification MIL-P-11268:

General requirements for plastic material and plastic parts.
Flux and cleaning agents for soldering
Process for soldering.
Cleaning.
Riveting.

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Wiring and Cabling, including:
Slack
Protection
Clearance
Splicing
Connections, general
Grounding, general
Shielding on wire and cable.

General requirements for securing of parts.
Cleaning of welding.
Process for welding
Mechanical operation of controls
Gears
Self-tapping screws.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- Inspection shall be classified as follows:

(a) First article inspection (does not include preparation for delivery). (See 4.3)

(b) Inspection covered by subsidiary documents. (See 4.4)

(c) Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.4 and 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.13)

4.3 First article inspection.- This inspection shall consist of the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for Group A, Group B and Group C (See Tables III, IV, and V respectively). The Group C portion of the First Article inspection shall be performed in the order shown in Table II according to the number of samples specified in the contract or order.

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Table II - Order of First Article Inspection

Inspection	Unit 1	Unit 2	Unit 3	Unit 4	Req. Para	Insp Para.
Non-damaging						
Temperature					3.8.2	4.7.1
Low	1				3.8.2.1	4.7.1.1
High	2				3.8.2.2	4.7.1.2
Altitude (elevation)		1			3.8.4	4.7.3
Leakage (immersion)		2			3.8.5	4.7.4
Dust		3			3.8.6	4.7.5
Vibration (internal)		4			3.8.8.1	4.7.6
Potentially damaging						
Bounce		5			3.8.8.2	4.7.7
Shock, Ballistic		6			3.8.8.2	4.7.8
Interference		7			3.14	4.4
Humidity (moisture resistance)	3				3.8.3	4.7.2
Fungus			1 (Note 1)		3.16	*4.15
Salt Fog				1 (Note 1)	3.8.7	*4.16

NOTE 1: The equipment shall be thoroughly washed, cleaned, dried, and refurbished, if necessary, before proceeding with subsequent tests.

* THIS TEST SHALL BE PERFORMED ON A SEPARATE SAMPLE.

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4.4 Inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the acceptance inspection before preparation for delivery.

Item	Where Required
Printed wiring assemblies	3.4
Finish, protective	3.6
Marking	3.7
Electromagnetic interference	3.14

4.5 Quality conformance inspection of equipment before preparation for delivery.- Each unit of each lot of equipment shall be subjected to preconditioning (see 3.10) prior to inspection. The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.3.2. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. Failure to meet any requirement as a result of preconditioning shall be classified as a defect and recorded as such. The government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the government, at its discretion, may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements (see 6.7). Test equipment for government verification inspection shall be made available by the contractor.

4.5.1 Group A inspection.- Each unit of each lot of equipment shall be inspected for conformance to all the inspection and test requirements of Table III. Each lot will be subject to verification, utilizing the procedures of MIL-STD-105 using the general inspection levels and the AQL's indicated in Table III. Group A inspection shall be performed in any order which is satisfactory to the Government, except that the air-seal (4.11) shall be next to last and the operational inspection (4.12) shall be last (see 6.4).

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Table III - Group A Inspection

Inspection	Req. Para.	Insp Para.	AQL Major	AQL Minor
Visual and mechanical	3.17	4.10		
Circuit board assembly			0.65%	2.5%
AM-2060( )/GRC assembly (without cover)			1.0%	4.0%
AM-2060( )/GRC complete assembly			1.0%	4.0%
Electrical				
Output voltage	3.9.1	4.8.2)	1.0%	*
Input current	3.9.2	4.8.3)	for	
Audio output	3.9.3	4.8.4)	the group	
Audio distortion	3.9.4	4.8.5)	combined	
Antenna frequency selector control	3.9.5	4.8.6)		
Air seal	3.11	4.11	1.0%	
Operational	3.12	4.12	1.0%	

* All electrical defects are considered major.

4.5.2 Group B inspection.- This inspection, including sampling, shall conform to Table IV and to special procedures for small-sample inspection of MIL-STD-105. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection.

4.5.2.1 Group B sampling plans.- The Group B sampling plans, for the AQL's listed in Table IV, shall be as follows:

AQL	Inspection Level
4.0%	S-4
6.5%	S-4

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4.5.2.2 Order of inspection within Group B.- Group B inspection shall be performed in any order which is satisfactory to the government.

Table IV - Group B Inspection

Inspection	Req Para	Insp Para	AQL (See 4.5.2.1)
Electrical			
No load voltage	3.9.6	4.8.7)	4.0%
Audio response	3.9.7	4.8.8)	for the
Input voltage range	3.9.8	4.8.9)	group combined
Interchangeability (dimensions)	3.13	4.9	6.5%

4.5.3 Group C inspection.- This inspection shall consist of the tests specified in Table V and shall be performed on sample units that have been subjected to and met Group A and Group B inspection. Each sample unit used for the salt fog or fungus test need meet only those environmental Group B inspections indicated for that unit in Table II. Sample units shall be selected in accordance with paragraph 4.5.3.1.

Table V - Group C Inspection

Inspection	Req Para	Insp Para
Subgroup I (see 4.5.3.1.1)		
Leakage (immersion)	3.8.5	4.7.4
Temperature	3.8.2	4.7.1
Subgroup II (See 4.5.3.1.2)		
Vibration, internal	3.8.8.1	4.7.6
Bounce	3.8.8.2	4.7.7
Shock, ballistic	3.8.8.2	4.7.8
Humidity (moisture resistance)	3.8.3	4.7.2
Subgroup III (see 4.5.3.1.3)		
Altitude (elevation)	3.8.4	4.7.3
Dust	3.8.6	4.7.5
Interference	3.14	4.4
Fungus	3.16	*4.15
Salt fog	3.8.7	*4.16

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* This test shall be performed on a separate sample.

4.5.3.1 Sampling for inspection of equipment.- Units selected for each Group C inspection shall be selected without regard to their quality in accordance with the following:

4.5.3.1.1 Subgroup I.- For this subgroup, two units from the first 25 production units shall be selected for the first Group C inspection. For subsequent Group C inspections, two units from each successive 100, or two per month if less than 100 are produced monthly, shall be selected. For this subgroup, the maximum number of units to be tested in a single month shall be 10.

4.5.3.1.2 Subgroup II.- For this subgroup, two units from the first 25 production units shall be selected for the first group C inspection. For subsequent Group C inspection, two units every four (4) months shall be selected.

4.5.3.1.3 Subgroup III.- For this subgroup two units from the first 25 production units shall be inspected for the first Group C inspection. No subsequent inspection for the tests in this subgroup will be required.

4.5.3.2 Noncompliance.- The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (1) the process and materials as applicable, and (2) all units of product which were manufactured under the same conditions and which the government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the government) and Groups A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective. (See 6.5).

4.5.4 Reinspection of conforming Group C sample units.- Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract or order provided all damage is repaired and the sample units are resubjected to and pass group A inspection.

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4.6 Preconditioning.- The preconditioning tests of paragraphs 4.6.1 and 4.6.2 below shall precede all other electrical tests of the equipment. Bounce preconditioning (4.6.1) shall precede electrical preconditioning (4.6.2)

4.6.1 Bounce preconditioning.- The amplifier-power supply shall be placed in its normal operating position on the table of the Package Tester, Type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York, or equal. Accessories shall be selected from those listed on Drawing SC-A-46439. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. Each unit of each lot of equipment shall be subjected to this bounce preconditioning for 1 minute prior to selection of units for inspection. After bounce preconditioning the unit shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to inspection. (See 3.10.1)

4.6.2 Electrical preconditioning (See 3.10.2).- The equipment shall be operated for a period of 48 hours with the following temperature and vibration conditions:

Temperature: +40 ±5°C

Vibration: 2.2 G ± 10% peak acceleration value at any non-resonant frequency between 20 and 55 Hz measured at the mounting points on the equipment. The duration of vibration shall be at least 10 minutes during each hour of equipment operating time. Operational inspection (3.12) shall be performed at the beginning and end of the 48 hour preconditioning. After initial measurements have been made, the equipment shall be operated with a transmit to receive ratio of 1 hour in transmit mode and 1 hour in receive mode. Monitor Output voltage (3.9.1), Input current (3.9.2), Audio output (3.9.3) and Antenna Frequency control (3.9.5) after every twelve (12) hours of conditioning. If the equipment fails any test during the 48 hours preconditioning, the cause of failure shall be corrected and the 48 hour preconditioning continued. The last 24 hours of conditioning shall be failure free.

NOTE: Equipment with component failures encountered during Group A, B or C testing which require component replacement need not be subjected to recycling through electrical preconditioning if the repair is made at the component level. If higher levels of assemblies are used to correct or repair component failures, the equipment must be recycled through electrical preconditioning or the replacement assembly must have previously been subjected to electrical preconditioning. Documentation and storage of preconditioned replacement assemblies must be maintained by the contractor.

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4.7 Service conditions tests.- The bounce and shock tests shall be performed with a load simulating Receiver-Transmitter, Radio RT-505( )/PRC-25 or RT-841( )/PRC-77 mounted on Amplifier-Power Supply AM-2060( )/GRC. The loads shall be approved by the government.

4.7.1 Temperature test.-

4.7.1.1 Low temperature.- The equipment shall be tested in accordance with Method 502, Procedure I, of MIL-STD-810. The storage temperature shall be -80°F and the operating temperature shall be -40°F. Prior to testing the equipment shall meet full specification performance for the measurements specified in 3.8.2.1.

4.7.1.2 High temperature.- The equipment shall be tested in accordance with Method 501, Procedure II, of MIL-STD-810. The storage temperature shall be 160°F and the operating temperature shall be 150°F. Prior to testing the equipment shall meet full specification performance for the measurements specified in 3.8.2.2

4.7.2 Humidity (moisture-resistance) test.- The equipment shall be tested in accordance with Method 507, Procedure II of MIL-STD-810. Prior to testing the equipment shall meet full specification performance for the measurements specified in 3.8.3. Measurements shall be made during the periods shown on the figure for the cycle. Do not move equipment from the humidity chamber for measurements, and do not leave power on after measurements have been completed. If any equipment fails to meet the specified performance during cycling, it shall be realigned or readjusted once and the test continued. No repair or replacement of parts shall be permitted. If the equipment still fails to meet the specified performance during cycling, or fails subsequently, it does not pass the test. In addition, if the equipment fails to meet specified requirements after final conditioning and adjustment, it does not pass the test.

4.7.3 Altitude test.- The equipment shall be tested in accordance with Method 500, Procedure I of MIL-STD-810. Prior to testing the equipment shall meet full specification performance for the measurement specified in 3.8.4.

4.7.4 Leakage (immersion).- The equipment, as prepared for field transportation, shall be tested in accordance with Method 512, Procedure I of MIL-STD-810 (See 3.8.5).

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4.7.5 Dust test.- The equipment shall be tested in accordance with Method 510, Procedure I of MIL-STD-810. Prior to testing the equipment shall meet full specification performance for the measurement specified in 3.8.6.

4.7.6 Vibration (Resonance search).- The equipment shall be tested in accordance with Method 514, Procedure IX, Part I, of MIL-STD-810 to determine conformance to 3.8.8.1.

4.7.7 Bounce test.- The equipment shall be tested in accordance with Method 514, Procedure IX, Part 2, of MIL-STD-810. The equipment shall be energized near the end of each 3/4 hour period and an operational or talk test performed. At the completion of the bounce test, the equipment shall be tested for compliance with 3.8.8.2.

4.7.8 Shock test; ballistic.- The test shall be conducted on the "Shock Testing Machine for Light Weight Equipment" shown in Specification MIL-S-901. The equipment, mounted on Mounting MT-1029( )/VRC, shall be secured in its normal operating position to the steel test plate by means of the same fasteners used for vehicular installation of the equipment. The test shall consist of a total of 9 blows: one each 1-foot blow, 3-foot blow, and 5-foot blow on the back, side, and top of the test plate. As an alternative to reorient the test plate for the blows on the side of the plate, equivalent rotation of the equipment under test is permissible. At the completion of the ballistic shock test the equipment shall be tested for compliance with 3.8.8.2.

4.8 Electrical tests, Amplifier-Power Supply AM-2060( )/GRC.-

4.8.1 Standard test conditions.- All electrical tests shall be conducted under the following conditions, unless otherwise specified for the individual tests.

Temperature.- Room ambient

Humidit .- Room ambient

Input voltage.- 26.0 vdc nominal

Load resistors.- Output load resistors simulating Receiver-Transmitter, Radio RT-505( )/PRC-25 or RT-841(.)/PRC-77 shall be utilized for electrical testing unless otherwise specified. The output loads shall be as follows:

Mode	Resistance (ohms)	
	B1	A1
Receive, Squelch-on	302 +0 -5%	_____

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Mode	Resistance (ohms) B1	A1
Transmit, Squelch-on	$14.5^{-5\%}_{+0}$	7.0 ±1.0% for 22.0 vdc input
	•	7.8 ±1.0% for 26.0 vdc input
		8.0 ±1.0% for 30.0 vdc input

4.8.2 Output voltage (see 3.9.1).- The amplifier-power supply shall be tested for voltage output using a voltmeter with a resistance of 1000 ohms per volt and an accuracy tolerance of not more than one percent. The power supply shall meet the requirements of 3.9.1.

4.8.3 Input current (see 3.9.2).- During the performance of the test of 4.8.2 the input current shall be measured to determine compliance with 3.9.2.

4.8.4 Audio output (see 3.9.3).- With the audio input to the amplifier adjusted to 3 milliwatts at 1000 H_z, the audio output shall be measured to determine compliance with 3.9.3.

4.8.5 Audio distortion (see 3.9.4).- The input signal to the amplifier shall be adjusted to obtain one watt output at 1000 H_z. The distortion output shall be measured to determine compliance with 3.9.4.

4.8.6 Antenna frequency selector control (see 3.9.5).- Amplifier-Power Supply AM-2060( )/GRC shall be operated from a 26.0 vdc source. For each of the ten control positions listed below, the voltage measured across the pins of the antenna control connector J4, shall meet the requirements of 3.9.5.

Switch Position		Voltage at pins
1	(30-33 MHz)	A to C, D to C, and N to C
2	(33-37 MHz)	A to C, E to C, and N to C
3	(37-42 MHz)	A to C, F to C, and N to C
4	(42-47 MHz)	A to C, H to C, and N to C
5	(47.5-53 MHz)	A to C, J to C, and N to C
6	(53-56 MHz)	B to C, D to C, and N to C
7	(56-60 MHz)	B to C, E to C, and N to C
8	(60-65 MHz)	B to C, F to C, and N to C
9	(65-70.5 MHz)	B to C, H to C, and N to C
10	(70.5-76 MHz)	B to C, J to C, and N to C

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4.8.7 No load voltage (see 3.9.6).- With 30.0 vdc applied to the amplifier-power supply, the no load voltage under both receive and receive-transmit conditions shall be measured to determine compliance with 3.9.6.

4.8.8 Audio response (see 3.9.7).- The input signal to the amplifier shall be adjusted to obtain 1 watt output at a frequency of 1000 Hz. With this input signal, the audio frequency response shall be measured at frequencies of 150, 300, 2500, 3000, and 5000 Hz to determine compliance with 3.9.7.

4.8.9 Input voltage range (see 3.9.8).- To determine compliance with 3.9.8, the tests listed below shall be conducted. At each of the input voltages, the set power shall be turned on and off 5 times.

Output voltage	4.8.2
Input current	4.8.3
Audio output	4.8.4

4.9 Inspection for dimensional interchangeability.- The dimensions of the equipment shall be gaged or measured in accordance with Electronics Command Drawing SG-GL-57992 to determine conformance to the physical interchangeability requirement of 3.13. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

4.10 Visual and mechanical inspection.- The units shall be examined for the applicable defects listed in MIL-STD-252.

4.11 Air seal test (see 3.11).- The equipment shall be opened and closed again in such a manner as to break and remake the seals. Immediately thereafter, the equipment, as field transported, shall be subjected to a vacuum of one pound per square inch (one pound per square inch less than the atmospheric pressure surrounding the equipment) applied to the interior of the equipment enclosure. The vacuum shall be valved off and interior pressure measured during the ensuing period of 1 minute. During this 1 minute period the decrease in vacuum shall not exceed .01 pound. The gage used for measurement of the vacuum shall be of such accuracy that a difference of .01 pound can be determined readily.

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4.12 Operational inspection (see 3.12).- Amplifier-Power Supply AM-2060( )/GRC shall be operated at nominal input voltage and ambient temperature and humidity. A listening test to determine proper operation of the amplifier and loudspeaker shall be conducted. In addition, the power supply output voltages shall be measured in both the receive and transmit mode.

4.13 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5.

4.14 Rough handling test(preparation for delivery) (see 6.2(d)).- When rough handling test in accordance with Specification MIL-P-116 is required by the contract, the following functional test shall be conducted to determine freedom from operational malfunction caused by rough handling.

Test	Paragraph
Operational inspection	4.12 (Listening test only)

4.15 Fungus test.- The equipment (including accessories) shall be tested in accordance with Method 508, Procedure I, of MIL-STD-810. Spraying shall include parts and materials on and under the chassis, and the interior of the equipment case. The equipment shall remain in an open condition during the test. Operation is not required. (See 3.16)

4.16 Salt fog test.- The equipment (including accessories) in its normal operating position shall be tested in accordance with Method 509, Procedure I, of MIL-STD-810. The equipment shall not be opened. Operation is not required. No rust shall be imposed on the test item by the test facility. (See 3.8.7).

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging of Amplifier Power Supply Group OA-3633( )/GRC shall be Level A or C as specified. (See 6.2(c)).

5.1.1 Level A.-

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5.1.1.1 Cleaning.- Amplifier-Power Supply Group shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying.- Amplifier-Power Supply Group shall be dried in accordance with the applicable procedures of MIL-P-116.

5.1.1.3 Preservation applications.- None required.

5.1.1.4 Unit packaging.- Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 as specified herein.

5.1.1.4.1 Technical literature.- Each technical manual shall be packaged method 1C-1.

5.1.1.4.2 Amplifier-Power Supply Group OA-3633( )/GRC.- Each amplifier-power supply group shall be packaged in accordance with items 1 through 10 of Table VI and Figure 1.

5.1.2 Level C.- Each amplifier-power supply group shall be preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration and damage during shipment from the supply source of the first receiving activity.

5.2 Packing.- Packing of Amplifier-Power Supply Group OA-3633( )/GRC shall be Level A, B or C as specified. (See 6.2(c)).

5.2.1 Level A.-

5.2.1.1 Palletized Load.- A quantity of Amplifier-Power Supply groups, packaged as specified in 5.1, shall be placed on a pallet, load type I, conforming to MIL-STD-147. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least twelve inches to accomodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant W5s. The load shall be bonded to the pallet by strapping.

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5.2.1.2 Less than palletized load.- When quantities per destination are less than a pallet load, a quantity of items packaged as specified in 5.1, shall be packed within a close-fitting fiberboard box conforming to PPP-B-640, Class 2, Style E. or MIL-B-43291. The gross weight of a box conforming to PPP-B-640 shall not exceed 200 pounds. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by nonmetallic strapping or pressure sensitive filament tape conforming to PPP-T-76, Type II or PPP-T-97, Type IV respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification.

5.2.2 Level B.- Each Amplifier-Power Supply Group, packaged as specified in 5.1, shall be packed as specified in 5.2.1.

5.2.3 Level C.- Each Amplifier-Power Supply Group, packaged as specified in 5.1, shall be packed as specified in 5.2.1 except that pallets, fiberboard boxes and fiberboard caps shall be of the type, size and kind commonly used for the purpose and shall comply with the rules and regulations of the common carrier as applicable to the mode of transportation. Packing shall be accomplished in a manner that will afford adequate protection to the package and its contents during shipment from the supply source to the first receiving activity.

5.3 Marking.- In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- Amplifier-Power Supply Group OA-3633( )/GRC, when installed on Mounting MT-1029( )/VRC, provides the dc power, mounting provisions, and antenna information required for vehicular operation of Receiver-Transmitter, Radio RT-505( )/PRC-25 and RT-841( )/PRC-77 when used as part of Radio Set AN/VRC-53( ), AN/VRC-64, AN/GRC-125( ) and AN/GRC-160( ).

6.2 Ordering data.- Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Type required.

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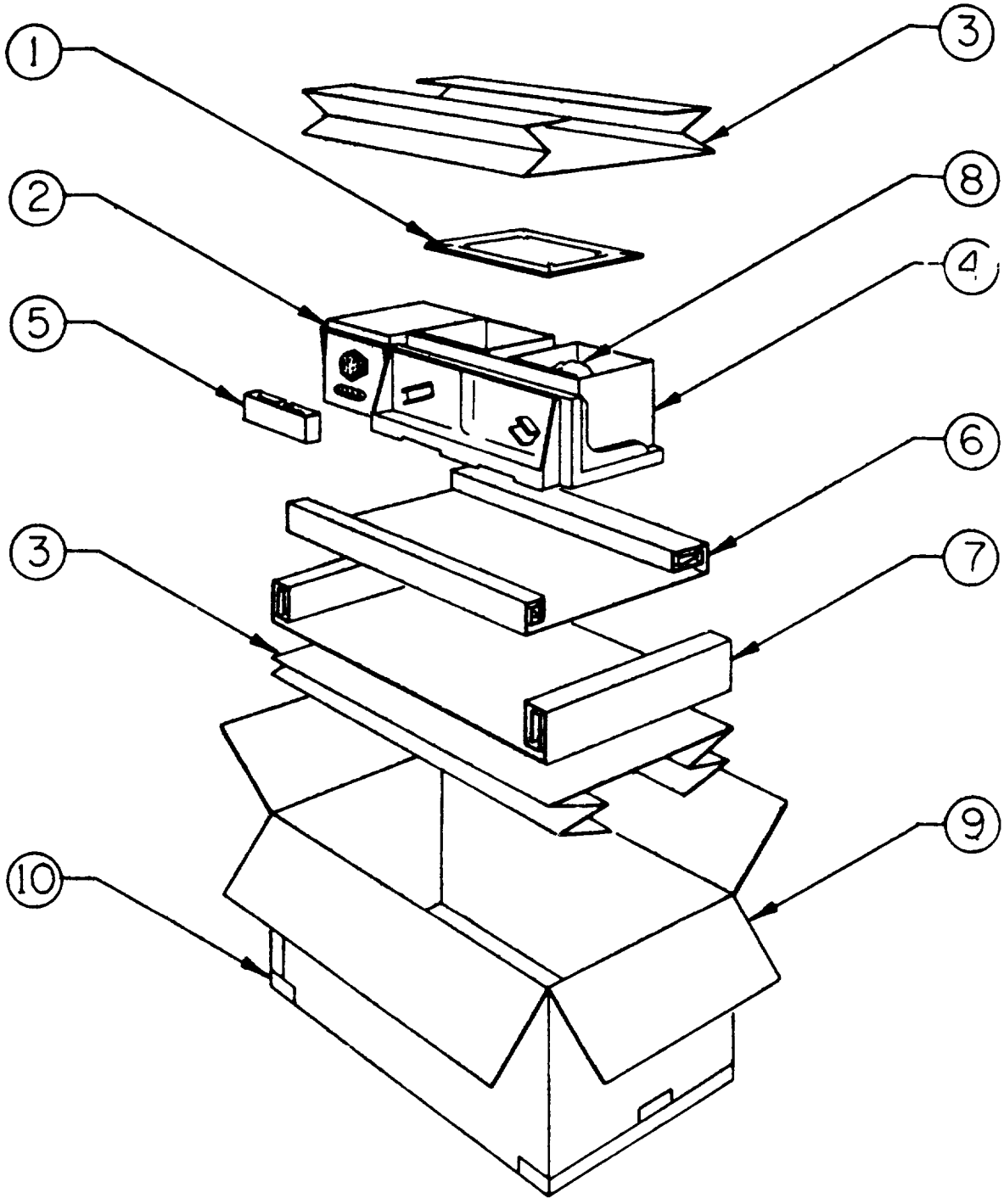


FIGURE 1

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BILL OF MATERIAL
TABLE VI

REF No.	ITEM Nomenclature	No. Req'd	Method MIL-P-116	SIZE - INCHES L	W	D	REQUIREMENT Spec No.	T	C	V	G	F	S
1	TECH MANUAL	2											
	Bag, Barrier	2	1C-1	12	9		MIL-B-117	11	C				
2	Amplifier Power Supply AM-2060()/GRC	1	111	15	13	6-5/8							
3	PAD, TOP and Bottom	2		17	14	3/4	PPP-F-320	CF	WR	SW	W5c		
4	CELL	1		10	9	5	PPP-F-320	CF	WR	SW	W5c		
5	CELL	1		5	2	2	PPP-F-320	CF	WR	SW	W5c		
6	CELL	1		15	14	2	PPP-F-320	CF	WR	SW	W5c		
7	CELL	1		17	14	4	PPP-F-320	CF	WR	SW	W5c		
8	CABLE ASSY CX-4655 Bag, Barrier	1	1C-1	14	6		MIL-B-121	1	1		A		
9	Box, Fiberboard	1		17-1/4	15	8-3/4	PPP-B-636	CF	WR	SW	W5c		RSC
10	TAPE	2		23	3		PPP-T-76						
		4		21	3								
		1		14-3/4	3								

Lev. A - PACKED WEIGHT
Lev. A - PACKED VOLUME

T - Type
C - Class
V - Variety
G - Grade
F - Flute
S - Style

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c. Level of preservation, packaging and packing. (See Section 5)

d. When the rough handling and functional tests are required. (See 4.14)

e. Marking and shipping of samples.

f. Place of final inspection.

g. Technical literature required. (See 3.15)

6.3 Nomenclature.- The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design, for example: AM-2060W/GRC. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1)

6.4 Location of operational inspection and air-seal test.- It is desirable that the operational inspection (4.12) and the air-seal test (4.11) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is complete. Any preparation for shipment which would require breaking of the equipment seal should be accomplished prior to the air-seal test so that the seal may remain intact thereafter. It is recommended that the entire lot (including all previously inspected sample units) be sampled immediately prior to packaging.

6.5 Group C inspection.- Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.2)

6.6 Color.- The color chip furnished by the procuring agency (3.6) will match color chip No. X-24087 and may be obtained upon request to Commanding General, US Army Electronics Command, ATTN: AMSEL-PP, 225 South Eighteenth Street, Philadelphia, Pennsylvania 19103.

6.7 Verification inspection.- Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

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6.8 Dimensional data.- Sizes of packaging materials prescribed in Section 5 are based on the dimensions of the equipment cited on the applicable Bill of Material. When the dimensions of the equipment vary from those cited, the sizes of the packaging materials shall be adjusted accordingly. When shown in the Bill of Materials (See Section 5), corrugated fiberboard manufactured with A, B or C fluting may be used at the option of the contractor. When the fluting used is not the same as that cited in the Bill of Material, the dimensions of the affected packaging and packing materials will be adjusted accordingly.

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Army - EL

Project Number 5820-A492

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

ESC-FM 1068-68

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